Dot Product and Geometry, $\vec{\S}$6.1, 6.2

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that
$$Av = \lambda v.$$
- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \ldots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \ldots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.
- **Theorem 7:** Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \ldots, \lambda_p$.
    1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k.
    2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
    3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k, then $\mathcal{B}_1, \ldots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

Definition 1. *Let $u, v \in \mathbb{R}^n$, then the* **dot product** *of u and v is given by*

$$u \cdot v = u^T v = u_1 v_1 + \cdots + u_n v_n.$$

Theorem 6.1: Let $u, v, w \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

1. $$u \cdot v = v \cdot u.$$

2. $$(u + v) \cdot w = u \cdot w + v \cdot w.$$

3. $$(cu) \cdot v = c(u \cdot v) = u \cdot (cv).$$

4. $u \cdot u \geq 0$ and $u \cdot u = 0$ if and only if $u = 0$.

Definition 2. *The* **length** *or* **norm** *of the vector v is a non-negative scalar, denoted by $\|v\|$, and defined by*

$$\|v\| = \sqrt{v \cdot v}.$$

Definition 3. *We define the* **distance** *between u and v by*

$$dist(u, v) = \|u - v\|.$$

Definition 4. *The vectors u and v are* **orthogonal** *if $u \cdot v = 0$.*

Theorem 6.2: The vectors u and v are orthongal if and only if

$$\|u\|^2 + \|v\|^2 = \|u + v\|^2.$$

1. Prove the Theorem.

Definition 5. *The set*

$$W^{\perp} = \{z \in \mathbb{R}^n : z \cdot w = 0 \text{ for all } w \in W\}$$

is called the **orthogonal complement** *of W.*

Note: $x \in W^{\perp}$ if and only if x is orthogonal to each vector in a set that spans W. Also $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Theorem 6.3 Let A be an $m \times n$ matrix, then

$$(\text{Row}(A))^{\perp} = \text{Null}(A) \text{ and } (\text{Col}(A))^{\perp} = \text{Null}(A^T).$$

2. Prove the Theorem.

Cauchy-Schwarz Inequality: For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|.$$

1. Prove Cauchy-Schwarz.

Note: We can use the dot product to find the angles between vectors; namely

$$u \cdot v = \|u\| \, \|v\| \cos\theta.$$

Triangle Inequality: For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ we have $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

2. Prove the triangle inequality.

Definition 6. *A set of vectors* $\{u_1, \ldots, u_p\}$ *in* $\mathbb{R}^n$ *is* **orthogonal** *if* $u_k \cdot u_j = 0$ *for all* $j \neq k$.

Theorem 6.4: If $S = \{u_1, \ldots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for Span S.

3. Prove the Theorem.

Definition 7. *An* **orthogonal basis** *for a subspace* W *of* $\mathbb{R}^n$ *is a basis for* W *that is also an orthogonal set.*

Theorem 6.5: Let $\{u_1, \ldots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \cdots + c_p u_p$$

where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

4. Prove the Theorem.

5. Let $u_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ and $x = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$. Find x in the $\{u_1, u_2\}$-coordinates.